

# Sales Prediction with Python

Predicting product sales from advertising spend across TV, Radio, and Newspaper channels using multiple regression models — with full EDA, residual analysis, and business insights.

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## Problem Statement

Marketing teams need to know **which advertising channels drive the most sales** and by how much. This project builds predictive models to estimate sales based on advertising budgets, helping businesses optimise their media spend for maximum ROI.

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## Dataset

- **Source:** Advertising Dataset — Kaggle
- **Size:** 200 records × 4 features
- **Target Variable:** Sales (in \$K)

| Feature          | Description                                   |
|------------------|-----------------------------------------------|
| <u>TV</u>        | TV advertising budget (\$K)                   |
| <u>Radio</u>     | Radio advertising budget (\$K)                |
| <u>Newspaper</u> | Newspaper advertising budget (\$K)            |
| <u>Sales</u>     | Product sales generated (\$K) — <b>Target</b> |

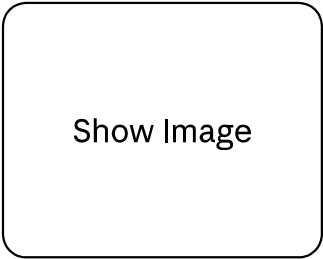
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## Visualisations

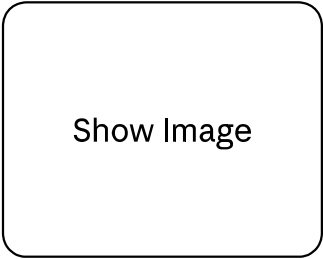
### 1. Advertising Channel Spend vs Sales

Show Image

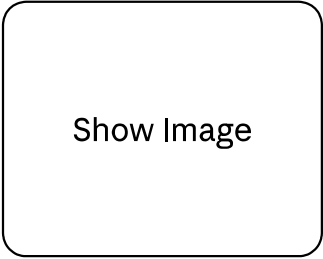
**2. Feature Correlation Heatmap**



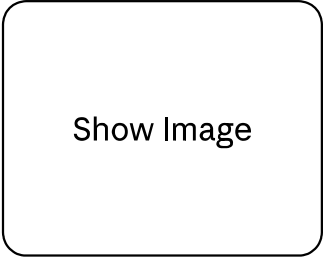
**3. Actual vs Predicted Sales — Linear Regression**



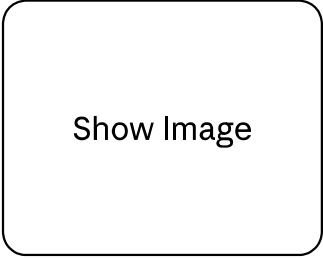
**4. Model Performance Comparison**



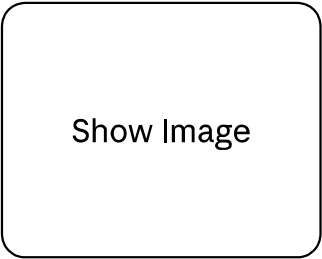
**5. Advertising Channel Impact (Coefficients)**



**6. Residual Analysis**



## 7. Budget Distribution vs Sales Contribution



### Models & Results

| Model             | R <sup>2</sup> Score | MAE (\$K) | RMSE (\$K) |
|-------------------|----------------------|-----------|------------|
| Linear Regression | 0.899                | 1.461     | 1.782      |
| Ridge Regression  | 0.899                | 1.461     | 1.782      |
| Lasso Regression  | 0.900                | 1.460     | 1.781      |
| Random Forest     | 0.981                | 0.620     | 0.769      |
| Gradient Boosting | 0.983                | 0.619     | 0.730      |

 **Best Model:** Gradient Boosting Regressor with R<sup>2</sup> = **0.983**

### Business Insights

-  **TV is the strongest driver** — highest coefficient weight; every \$1K in TV budget yields the most sales uplift.
-  **Radio has strong ROI** — second most impactful channel with a solid contribution to sales.
-  **Newspaper shows low influence** — despite receiving a significant share of budget, it contributes the least to sales.
-  **Ensemble models dominate** — Random Forest and Gradient Boosting far outperform linear models, confirming non-linear interactions between channels.
-  **Optimised allocation matters** — a \$200K TV + \$40K Radio + \$50K Newspaper campaign predicts ~\$19.6K in sales.

## Tech Stack

| Tool                      | Purpose                           |
|---------------------------|-----------------------------------|
| <code>pandas</code>       | Data loading and preprocessing    |
| <code>numpy</code>        | Numerical operations              |
| <code>scikit-learn</code> | Model training, evaluation        |
| <code>matplotlib</code>   | Custom dark-themed visualisations |
| <code>seaborn</code>      | Heatmaps and distribution plots   |

## How to Run

```
bash

# 1. Clone the repo
git clone https://github.com/<your-username>/sales-prediction-python.git
cd sales-prediction-python

# 2. Install dependencies
pip install pandas numpy matplotlib seaborn scikit-learn

# 3. Place the dataset in the root directory
#   - Advertising.csv

# 4. Run the analysis
python sales_prediction.py
```

All 7 plots will be saved to the working directory.

## Project Structure

```
sales-prediction-python/
|
├─ sales_prediction.py          # Main ML + EDA script
├─ Sales_Prediction_using_Python.ipynb  # Original notebook
```

```
├─ Advertising.csv          # Dataset
├─ plots/
│   ├─ plot1_channel_vs_sales.png
│   ├─ plot2_correlation_heatmap.png
│   ├─ plot3_actual_vs_predicted.png
│   ├─ plot4_model_comparison.png
│   ├─ plot5_channel_impact.png
│   ├─ plot6_residual_analysis.png
│   └─ plot7_budget_vs_impact.png
└─ README.md
```

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## 🍷 Connect

**Brahmini Seelam** — CSE Undergraduate @ AUCE, Visakhapatnam

🔗 [LinkedIn](#) · [GitHub](#)

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*This project was completed as part of an internship task at Oasis Infobyte.*